1.(a)	Sets $\frac{1}{2}r^2 \times 1.25 = 15 \Rightarrow r^2 = 24$	M1
	$\Rightarrow r = \sqrt{24}$ or $2\sqrt{6}$ (only)	A1
		(2)
(b)	Attempts $s = r\theta = 2\sqrt{6} \times 1.25$	M1
	Attempts $P = 2r + r\theta = 2 \times 2\sqrt{6} + 2\sqrt{6} \times 1.25$	dM1
	$= \frac{13\sqrt{6}}{2}$ oe	A1
		(3)
		(5 marks)

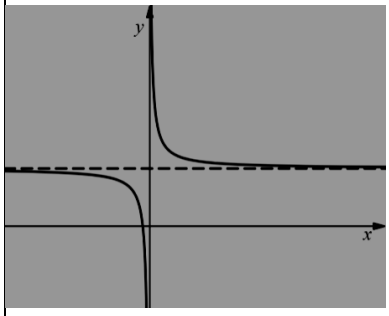
2. (a)	$1.85 = 2a + b$ and $3.45 = 7a + b$ Solves simultaneously to get $a = 0.32, b = 1.21$ (oe)	M1 A1 dM1 A1 **(4)**
(b)	States 1.21 m or 121 cm (oe)	B1ft **(1)** **(5 marks)**

3. (a)	$x^2-5x+13=\left(x-2.5\right)^2-2.5^2+13=\left(x-2.5\right)^2+6.75$	M1 A1
	Coordinates $M=\left(2.5,6.75\right)$	A1
		(3)
(b)	Attempts the equation of l using their M $y=\dfrac{6.75}{2.5}x$ $\left(y=2.7x\right)$	M1
	Attempts to solve their $y=2.7x$ with $y=x^2-5x+13$	
	$\Rightarrow 2.7x=x^2-5x+13\Rightarrow x^2-7.7x+13=0\ \Rightarrow\left(x-2.5\right)\left(x-5.2\right)=0$	M1
	$x=5.2$ oe	A1
	Coordinates $N=\left(5.2,14.04\right)$	dM1 A1
		(5)
(c)	States two of $y<x^2-5x+13,\ y>2.7x,\ 0\leqslant x<2.5$	M1
	States all three of $y<x^2-5x+13,\ y>2.7x,\ 0\leqslant x<2.5$	A1ft
		(2)
		(10 marks)

4. (a)	Area $ABCD$ is 40 cm^2 $\Rightarrow 40 = 6\times 10\times \sin\theta$ oe	M1
	$\sin\theta = \frac{2}{3} \Rightarrow \theta = 180° - 41.8°$	M1
	$\angle DAB =$ awrt $138.19°$	A1
		(3)
(b)	Attempts $DB^2 = 10^2 + 6^2 - 2\times 10\times 6\cos"138.19°"$	M1
	$DB =$ awrt 15.0 (cm)	A1
		(2)
		(5 marks)

5.(a)	$\frac{dy}{dx} = \frac{1}{2}x^2 + 2x^{-\frac{1}{2}}$	M1A1 A1 **(3)**
(b)	$\left.\frac{dy}{dx}\right\|_{x=4} = \frac{1}{2}\times 4^2 + 2\times\frac{1}{\sqrt{4}} = (9)$	M1
	Gradient of normal is $-\frac{1}{9}$	dM1
	$y - \frac{11}{3} = -\frac{1}{9}(x-4) \Rightarrow x + 9y - 37 = 0$	M1 A1 **(4)** **(7 marks)**

6. (a)



Shape — B1

States asymptote as $y = k$ — B1

States intercept as $-\frac{4}{k}$ — B1

(3)

(b)

$10 - 2x = \frac{4}{x} + k \Rightarrow 10x - 2x^2 = 4 + kx$ — M1

$\Rightarrow 2x^2 + (k-10)x + 4 = 0$ — A1

Attempts $"b^2 - 4ac" = 0 \Rightarrow (k-10)^2 - 4 \times 2 \times 4 = 0$ — M1

$k = 10 \pm 4\sqrt{2}$ oe — M1 A1

(5)

(8 marks)

7.(a)	Attempts $\frac{dy}{dx} = 4x$ at $x = 2$	M1
	At $x = 2$ gradient of tangent = 8	A1
		(2)
(b)	$\left(y_Q =\right)\ 2(2+h)^2 + 5$	B1
	Gradient $PQ = \frac{\text{their } y_Q - 13}{2+h-2}$	M1
	$\left(= \frac{8h + 2h^2}{h}\right) = 8 + 2h$	A1
		(3)
(c)	States as $h \to 0$ Gradient $PQ \to 8$ = Gradient of tangent	B1
		(1)
		(6 marks)

8	$x - 6x^{\frac{1}{2}} + 4 = 0$ $x^{\frac{1}{2}} = 3 \pm \sqrt{5}$ oe $x = \left(3 \pm \sqrt{5}\right)^2 \Rightarrow x = 14 \pm 6\sqrt{5}$	 M1 A1 M1 A1 A1 **(5 marks)**

8 Alt	$x+4=6x^{\frac{1}{2}}$ $(x+4)^2=36x$ $x^2-28x+16=0\Rightarrow(x-14)^2=180\Rightarrow x=14\pm\sqrt{180}\Rightarrow x=14\pm6\sqrt{5}$	 M1 A1 M1 A1 A1 **(5)**

9. (a)	24π	B1 **(1)**
(b)	$(18\pi, -1)$	B1ft **(1)**
(c)(i)	$-12\pi - \alpha$	B1 ft
(ii)	$6\pi - \alpha$	B1 ft **(2)** **(4 marks)**

10 (a)	$f(x) \leqslant 0 \Rightarrow \quad x \leqslant -\frac{5}{2}, \quad x = 3$	M1 A1

		(2)
(b)	$f(x)=(2x+5)(x-3)^2=(2x+5)(x^2-6x+9)$	M1
	$=2x^3-12x^2+18x+5x^2-30x+45$	M1
	$=2x^3-7x^2-12x+45$	A1
		(3)
(c)	(i) $P(0,45)$	B1ft
	(ii) Gradient = -12	B1ft
		(2)
(d)	(i) $g(x)=(2(x-2)+5)(x-2-3)^2=(2x+1)(x-5)^2$	M1 A1
	(ii) 25	B1
		(3)
		(10 marks)

11. (a)	Attempts $y-\frac{32}{3}=5(x-4)\Rightarrow y=5x-\frac{28}{3}$	M1 A1 **(2)**
(b)	$f''(x)=\frac{4}{\sqrt{x}}-3\Rightarrow f'(x)=8x^{\frac{1}{2}}-3x+k$	M1 A1
	Substitutes $x=4,\ f'(x)=5\Rightarrow k=1$	dM1 A1
	$f'(x)=8x^{\frac{1}{2}}-3x+1\Rightarrow f(x)=\frac{16}{3}x^{\frac{3}{2}}-\frac{3}{2}x^2+x+d$	dM1 A1
	Substitutes $x=4,\ f(x)=\frac{32}{3}\Rightarrow d=-12$	ddM1
	$f(x)=\frac{16}{3}x^{\frac{3}{2}}-\frac{3}{2}x^2+x-12$	A1 **(8)** **(10 marks)**